Gun-Woo Bae

Homepage | Tech Blog | Linkedin | Github

Education

Gwangju Institute of Science and Technology (GIST) **Feb. 2024 – Feb. 2026**
AIMED Lab of Prof. Mansu Kim *Gwangju, Korea*
Master of AI Convergence
• GPA: 3.91 / 4.5 (3.48/4.0)

Jeju National University **Feb. 2017 – Aug. 2023**
Bachelor of Computer Science and Statistics & History *Jeju, Korea*
• GPA: 4.22 / 4.5 (3.75/4.0)

Publications

† Equal contribution ∗ Corresponding author

[1] **Gunwoo Bae**†, Yeonwoo Kim†, Junha Bae, Mansu Kim∗, "ConnecToMind2: Inter-Subject fMRI Decoding via Whole-Brain Connectome-Guided Alignment", *MICCAI*, 2026. [Spotlight (∼5%)] (Link)

[2] Youngjoo Lee†, **Gunwoo Bae**†, H. Suh, J. Cha, J. Jung, Sehhoon Park∗, Mansu Kim∗, "A real-world Fitbit-derived dataset of activity, sleep, and heart rate with matched clinical factors in on-treatment lung cancer patients", *Scientific Data*, 2026. [IF: 7.2, Top 11.8%] (Link) (Dataset)

[3] **Gunwoo Bae**†, Yeonwoo Kim†, Mansu Kim∗, "ConnecToMind: Connectome-Aware fMRI Decoding for Visual Image Reconstruction", *MICCAI MLMI Workshop*, 2025. [Poster] (Link)

[4] **Gunwoo Bae**, Mansu Kim∗, "Image classification based on functional MRI signal", *Conference of the Korean Society of Medical and Biological Engineering*, 2024. [Oral, Bronze Award] (Link)

Presentations

MICCAI – Spotlight & Poster [1] **Sep. 2026**
MICCAI MLMI Workshop – Poster [3] **Sep. 2025**
Conference of the Korean Society of Medical and Biological Engineering – Oral [4] **Oct. 2024**

Research Experience

Neural Decoding from fMRI **Jul. 2024 – May 2026**

- Extended the region-level embedding approach [3] to a whole-brain fMRI-to-image model, adding a Connectome-guided Q-Former guided by structural connectivity. Showed that brain regions outside the visual network also contribute to decoding visual stimuli. [1]
- Built an fMRI-to-image model with region-level embeddings for visual cortical sub-regions, guided by resting-state functional connectivity. Showed that these embeddings preserve the functional roles of brain regions and improve reconstruction quality. [3]
- Removed resting-state activity as noise from task fMRI via PCA, improving image classification. Showed that task-irrelevant activity hinders interpreting brain signals. [4]

Wearable Health Monitoring with Samsung Medical Center **Oct. 2024 – May 2026**

- Collected 3-month activity, sleep, and heart rate data from ∼250 lung cancer patients, validated it against prior studies, and published it as an open dataset. [2] [Dataset]
- Built G-FITBIT, a patient monitoring platform with interactive visualizations for clinicians and personalized reports for patients, using Django, PostgreSQL, Airflow, and Docker. [2] [Platform]
- Conducting a clustering analysis of wearable data to identify patient subgroups linked to objective clinical conditions. *(Ongoing)*

Engineering Projects

AutoX: Cloud Platform for Resource Sharing **Jan. 2025 – Feb. 2026**
In collaboration with GIST AI Convergence Department

- **Purpose:** Build a cloud service platform that enables sharing of idle GPU and CPU resources across research labs to support AI training tasks.
- **Role:** Connected Kubernetes with the back-end system, deployed the web service on Kubernetes, and implemented automated job scheduling and management using Airflow.
- **Tech Stack:** Kubernetes, Airflow, Django REST Framework, PostgreSQL, Vue.js, Nginx
- **Link:** www.autox.dreamai.kr

Technical Skills

Languages: Python, R, C

AI Tools: PyTorch, NumPy, Pandas

Brain Analysis Tools: FreeSurfer, ANTs, FSL, fMRIPrep

Server Tools: Kubernetes, Docker, CUDA, Airflow

Web Tools: Django, Django REST Framework, PostgreSQL

Scholarships & Awards

Best Paper Award (Bronze) **Oct. 2024**
Conference of the Korean Society of Medical and Biological Engineering

National scholarship student of Korea (Full Funding) **Mar. 2024 – Feb. 2026**
GIST

Merit-based scholarship (Top 10%, Half Funding) **Aug. 2020, Feb. 2022, Aug. 2022**
Jeju National University

Extracurricular Activities

Google Machine Learning Bootcamp
Google Korea **Sep. 2022 – Nov. 2022**

Military Service
Republic of Korea Army **July 2018 – Feb. 2020**